

Question 01

$$f(x) = x \ln(x)$$

①  $x = 1.0$ ,  $h = 0.1$

$$\begin{aligned} F.D &= \frac{f(x+h) - f(x)}{h} = \frac{f(1+0.1) - f(1)}{0.1} = \frac{f(1.1) - f(1)}{0.1} \\ &= \frac{1.1 \ln(1.1) - 1 \ln 1}{0.1} \end{aligned}$$

$$= 1.0484$$

$$\begin{aligned} C.D &= \frac{f(x+h) - f(x-h)}{2h} = \frac{f(1+0.1) - f(1-0.1)}{2 \times 0.1} = \frac{f(1.1) - f(0.9)}{0.2} \\ &= \frac{1.1 \ln(1.1) - 0.9 \ln(0.9)}{0.2} \end{aligned}$$

$$= 0.99833$$

②  $x = 1.0$ ;  $h = 0.1$

$$\Sigma [x-h, x+h]$$

$$\Sigma [0.9, 1.1]$$

$$f'(x) = \ln x + 1$$

$$f''(x) = \frac{1}{x}$$

$$f'''(x) = -\frac{1}{x^2}$$

The upper bound of truncation error,

$$B.D = \frac{f''(\xi)h}{2!} = \frac{0.1}{2x} = \frac{0.1}{2 \times 0.9} = 0.055555$$

$$C.D = \frac{f'''(\xi)}{3!} h^2 = \frac{(0.1)^2}{6x^2} = \frac{(0.1)^2}{6 \times 0.9^2} = 0.0020576$$

$$\textcircled{2} \quad D_h = f'(x) + \frac{f^3(x)}{3!} h^2 + \frac{f^5}{5!} h^4 + O(h^6)$$

$$D_{4h/3} = f'(x) + \frac{f^3(x)}{3!} \frac{16h^2}{9} + \frac{f^5(x)}{5!} \frac{256h^4}{81} + O(h^6)$$

$$\frac{9}{16} D_{4h/3} - D_h = -\frac{7}{16} f'(x) + O(h^4) + \left\{ \frac{f^5(x)}{5!} \frac{16h^4}{9} - \frac{f^5(x)}{5!} h^4 \right\} + O(h^6)$$

$$\Rightarrow \frac{\frac{9}{16} D_{4h/3} - D_h}{-7/16} = f'(x) + \frac{\left\{ \frac{f^5(x)}{5!} \frac{16h^4}{9} - \frac{f^5(x)}{5!} h^4 \right\}}{-7/16} + O(h^6)$$

$$\Rightarrow -\frac{9}{7} D_{4h/3} + \frac{16}{7} D_h = D_h^1$$

$$\therefore D_h^1 = \frac{16}{7} D_h - \frac{9}{7} D_{4h/3}$$

### Question 02

$$f(x) = x \cos(x) - x + \sin x$$

$$\text{let, } h = 0.1$$

$$C.D = \frac{f(x+h) - f(x-h)}{2h}$$

$$= \frac{f(1.2+0.1) - f(1.2-0.1)}{2 \times 0.1}$$

$$= \frac{f(1.3) - f(1.1)}{0.2}$$

$$= \frac{1.3 \cos(1.3) - 1.3 + \sin(1.3) - 1.1 \cos(1.1) + 1.1 - \sin(1.1)}{0.2}$$

$$= \frac{0.01131 - 0.2902}{0.2}$$

$$= -1.394$$

$$f(x) = x \cos x - x + \sin x$$

$$f'(x) = -x \sin x + \cos x - 1 + \cos(x)$$

$$= -x \sin x + 2 \cos x - 1$$

$$f'(1.2) = -1.2 \sin(1.2) + 2 \cos(1.2) - 1$$

$$= -1.393$$

$$\text{Relative error} = |-1.393 - (-1.394)|$$

$$= |1.394 - 1.393|$$

$$= 0.001000$$

Question 03

$$f(x) = 4x^3 - 9e^{7x}$$

①  $x = 2.7$

$$D_h' = \frac{4D_{h/2} - D_h}{3}$$

$$D_{(0.2)}^1 = \frac{4D_{0.1} - D_{0.2}}{3}$$

$$= \frac{4(-1102) - (-1383)}{3}$$

$$= -1008$$

$$D_{0.2} = \frac{f(2.7+0.2) - f(2.7-0.2)}{2 \times 0.2}$$

$$= -1383$$

$$D_{0.1} = \frac{f(2.7+0.1) - f(2.7-0.1)}{2 \times 0.1}$$

$$= -1102$$

$$\textcircled{b} \quad D_{0.2}^2 = \frac{16 D'_{h/2} - D'_h}{15}$$

$$= \frac{16 D'_{0.1} - D'_{0.2}}{15}$$

$$= \frac{16(-1016) - (-1008)}{15}$$

$$= -1016$$

$$D_{0.1}^1 = \frac{4 D_{0.05} - D_{0.1}}{3}$$

$$= \frac{4(-1038) - (-1102)}{3}$$

$$= -1016$$

$$D_{0.05} = \frac{f(2.7+0.05) - f(2.7-0.05)}{2 \times 0.05}$$

$$= -1038$$

$$D_{0.1} = -1102 \text{ [from 'a']}$$

$$D'_{0.2} = -1008 \text{ [from 'a']}$$